NASA ARTEMIS III REGIONS

13 Candidate Sites

Autonomous safety analysis evaluated across all 13 NASA Artemis III candidate landing regions using LRO LOLA elevation data and Horn slope topology algorithms.

NASA TOPOGRAPHY ASSETS

100% INTEGRATED

DEM Heightmap (80S_80M):	dem_heightmap.png
Horn Finite-Diff Slope:	slope_texture.png
RMS Regolith Roughness:	roughness_texture.png
Artemis III GeoJSON Catalog:	A3_Named_regions.geojson

LANDING SITE RECOMMENDATIONS

Awaiting terrain classification to extract contiguous safe landing sites.

Awaiting image analysis to generate cryptographic SHA-256 evidence chain.

MISSION TERMINAL (LIVE TELEMETRY)

7 EVENTS

```
03:38:33 [INFO] [NASA PDS]: Pinging NASA Planetary Data System Search API...
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03:38:45 [SUCCESS] [NASA PDS]: Connected (Latency: 12414ms, Found undefined lunar products)
03:38:46 [SUCCESS] [NASA PDS]: Connected (Latency: 13438ms, Found undefined lunar products)
```

Run AI Detection (External YOLO)

TEAM ROCKET · AEROX CLUB · PARUL UNIVERSITY

Scan History & Flight Recommendations

Project Head: **Darshan Patil** | Standard: NASA CLPS / Artemis III Criteria

Reset Store

No Scans Logged

Awaiting NASA LRO / LOLA DEM mission analysis to evaluate candidate landing zones.

ASK AGENT OM

NASA PLANETARY DATA CONNECTIONS

NASA PDS • LRO LOLA

NASA Planetary Data System (PDS Search API) — PING API

Endpoint:	pds.nasa.gov/api/search/1/products
Connection Status:	STANDBY
Query Latency:	13438 ms
Target Query:	q=(pds:Target_Identification.pds:name eq "Moon")

NASA LRO LOLA TOPOGRAPHY — 80S_80M DEM

Dataset:	LOLA GDR 80S_80M
Coverage:	80°S – 90°S (Lunar South Pole)
Resolution:	80 meters / pixel DEM
Source:	NASA PDS Geosciences

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NASA LRO South Pole (LOLA GDR 80S_80M)

AI Cross-Validation: YOLOv5 (Active) | Rank Matrix | 13 Artemis III Regions | ONLINE

NASA SENSOR CONTROLS

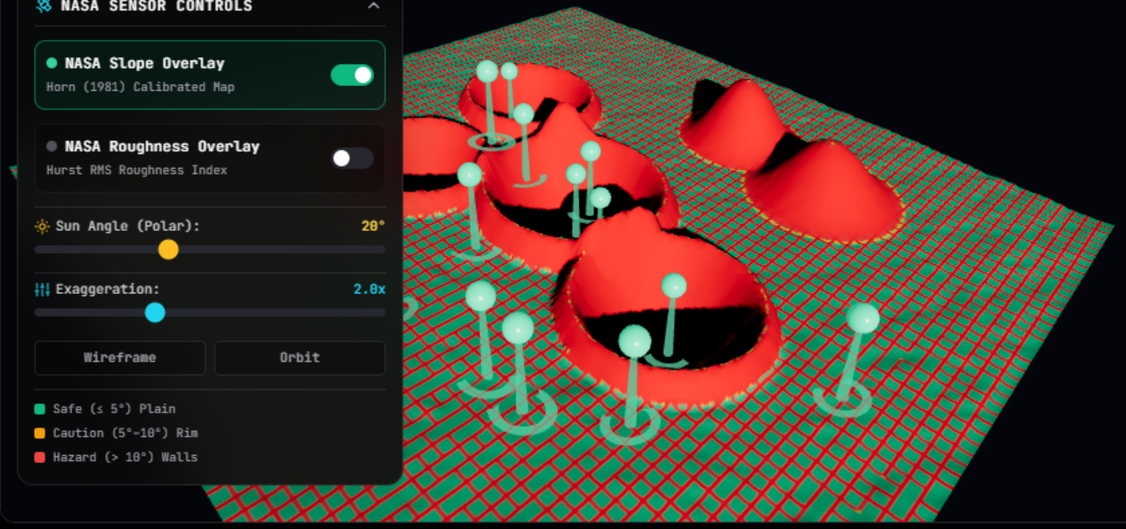
- **NASA Slope Overlay** — Horn (1981) Calibrated Map
- **NASA Roughness Overlay** — Hurst RMS Roughness Index

Sun Angle (Polar): 20°

Exaggeration: 2.0x

Wireframe | Orbit

- Safe (≤ 5°) Plain
- Caution (5°–10°) Rim
- Hazard (> 10°) Walls



LANDING SITE RECOMMENDATIONS

Awaiting terrain classification to extract contiguous safe landing sites.

PIPELINE MODULES & EXECUTION STATES

STRICT DETERMINISM

- GROUP 1: DETERMINISTIC CV ENGINE (ZERO AI CLAIM) — 0/0 EXECUTED
- GROUP 2: SCOPED GEMINI AI MODULES (NO NUMBER INVENTIONS) — 0/0 EXECUTED

AI-Assisted Mission Validation — GEMINI VISION

Validates authentic lunar surface morphology vs terrestrial imagery. Strictly does NOT calculate slopes or hazard stats.

Run Gemini Validation

AI Narrative Generator — GEMINI FLASH

Synthesizes already-computed deterministic CV numbers into a plain English landing briefing. Never invents or estimates numbers.

BRIEFING

Awaiting image analysis to generate cryptographic SHA-256 evidence chain.

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LANDING SITE RECOMMENDATIONS

Awaiting terrain classification to extract contiguous safe landing sites.

Awaiting

CLPS Intuitive Machines IM-1 (Odysseus)

SAFE TOUCHDOWN ZONE — PERMISSIBLE

LunarSafe Safety Score: 87 / 100

Measured slope (4.90°) is within the lander threshold (≤ 10.0°). Safe for footpad landing.

Supports Artemis Surface Infrastructure

Safe landing protects VIPER mobility and PSR water-ice harvesting.

Max Lander Slope Tolerance: 10.0°

8.0° (Strict NASA CLPS) | 10.0° (Standard) | 15.0° (High-Clearance)

LOLA SAMPLED TELEMETRY

Real Slope (NASA LRO): 4.90°	Elevation Datum: +2240 m
Roughness Index (LRO): 0.19	Solar Exposure: 91.5%
Coordinates: 85.99° S, 2.90° E	CLPS Mission: CLPS Intuitive Machi…

MISSION TERMINAL (LIVE TELEMETRY)

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03:38:46 [SUCCESS] [NASA PDS]: Connected (Latency: 13438ms, Found undefined lunar products)
03:39:10 [INFO] [NASA LRO Telemetry]: > Sampling NASA LRO slope at Malapert Mountain (85.99°S, 2.90°E)...
03:39:10 [SUCCESS] [CLPS Site Focus]: > Acquired Artemis III region: Malapert Mountain. Sampled Ground Slope: 4.90° (Tolerance: 10.0°)
```

Run AI Detection (External YOLO)

PRIMARY SCIENTIFIC TARGET

Peak of Eternal Light candidate / Telecoms Relay

MISSION & GEOLOGICAL CONTEXT

Historic landing site candidate for Intuitive Machines IM-1. Malapert Mountain stands over 5 km high with flat crest summits receiving over 90% annual illumination.

TEAM ROCKET · AEROX CLUB · PARUL UNIVERSITY

Scan History & Flight Recommendations

Project Head: **Darshan Patil** | Standard: NASA CLPS / Artemis III Criteria

No Scans Logged

Awaiting NASA LRO / LOLA DEM mission analysis to evaluate candidate landing zones.

LOLA SURFACE RELIEF PROFILE — NASA LRO (80S_80M)

NASA CLPS MISSION DECISION MATRIX | 100% DETERMINISTIC PHYSICS

Artemis III Candidate Sites Multi-Ranking

Comparative ranking across all 13 South Pole regions calibrated for lander tolerance: ≤ **10.0°**.

Total Evaluated Sites:	Permissible Touchdown Zones:	Top Touchdown Target:	Mean South Pole Slope:
13 Candidate Regions	**11 Safe / 2 Exceeded**	**Malapert Mountain**	**7.7°**

Search Artemis region or CLPS mission...

Export CSV

RANK	REGION NAME ↑↓	SAMPLED SLOPE ↑↓	SAFETY SCORE ↑↓	LIGHTING % ↑↓	ROUGHNESS	STATUS	ACTION
1	**Malapert Mountain** -85.99°S, 2.90°E • 2240m CLPS Intuitive Machines IM-1 (Odysseus)	4.9°	87	91.5%	0.19	SAFE	Active
2	**Connecting Ridge** -89.44°S, -137.10°E • 840m CLPS Nova-C / Lunar Trailblazer Relay	5.2°	86	82.0%	0.22	SAFE	Focus
3	**Leibnitz Beta Plateau** -85.83°S, 28.60°E • 1890m CLPS Commercial Cargo Transport	5.8°	84	85.0%	0.24	SAFE	Focus
#4	**Nobile Rim 1** -85.25°S, 53.50°E • 720m CLPS Astrobotic Griffin (VIPER Target)	6.1°	83	81.3%	0.26	SAFE	Focus
#5	**Haworth** -87.87°S, -5.10°E • 180m CLPS Micro-Nova Hopper	6.3°	82	71.0%	0.25	SAFE	Focus
#6	**Faustini Rim A** -87.16°S, 77.00°E • -240m CLPS PRISM / Polar Explorer	6.8°	81	74.5%	0.28	SAFE	Focus
#7	**Connecting Ridge Extension** -89.17°S, -155.60°E • 610m CLPS Masten / Astrobotic Lander	7.1°	80	78.4%	0.31	SAFE	Focus
#8	**Amundsen Rim** -84.50°S, 85.60°E • 580m CLPS Commercial Science Lander	7.4°	79	77.2%	0.30	SAFE	Focus
#9	**Nobile Rim 2** -85.38°S, 36.60°E • 650m CLPS Secondary Target Site	8.1°	76	79.7%	0.33	SAFE	Focus

FIPS 180-4 Deterministic Safety Assessment • NASA Artemis III Candidate Regions

Team Rocket • NASA Space Apps Challenge 2026

SHA-256 TELEMETRY VERIFICATION LEDGER SEALED

NASA Space Apps Challenge 2026 • Cryptographic Flight Safety Audit Trail

CANDIDATE REGION
Malapert Mountain

CLPS MISSION
CLPS Intuitive Machines IM-1 (Odysseus)

SAMPLED PHYSICAL SLOPE
4.90° (Tolerance: 10.0°)

LUNARSAFE SCORE
87 / 100 (SAFE)

IMMUTABLE SHA-256 HASH DIGEST

WebCrypto API • 256 bits

639dde7b789909c8531e38b450e718977b12374c118ca088823d356a5aa54b21

Canonical Audit Envelope (Hashed Payload)

Deterministic Key Ordering


```
{
  "auditType": "NASA_CLPS_LANDING_SITE_VERIFICATION",
  "siteName": "Malapert Mountain",
  "siteCode": "MALAPERT_MTN",
  "selenographicCoordinates": {
    "latitude": -85.99,
    "longitude": 2.9
  },
  "elevationMeters": 2240,
  "clpsMissionTarget": "CLPS Intuitive Machines IM-1 (Odysseus)",
  "measuredSlopeDeg": 4.9,
```


Cryptographic Non-Repudiation Guarantee: Any bit-level alteration in elevation data, candidate coordinates, or slope calculations produces a completely different SHA-256 digest, preventing retrospective falsification of landing hazard assessments.

NASA Planetary Data System (LOLA / LRO-L-LOLA-4-GDR-V1.0)

Dismiss